Reproducible Research Principles

Establishing trust, transparency, and continuity in science

ALASKA INBRE
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THE BEST OF

THE JOURNAL OF IRREPRODUCIBLE RESULTS

"Improbable Investigations & Unfounded Findings"

THE BEST OF THE PARODY MAGAZINE WRITTEN BY EMINENT MEMBERS OF THE SCIENTIFIC & MEDICAL COMMUNITY

Edited by
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INCLUDING:

Core Definitions

Reproducibility: Ability to recompute results given the same data set and the same analysis methods.

Replicability: Ability to produce a consistent result using different data but the same analysis methods.

Beware: Definitions may be flipped depending on who is talking!

		Data	
		Same	Different
Code	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Kirstie Whitaker (doi: 10.6084/m9.figshare.5440621)

Why These Principles Matter?

- Verify that your methodology is complete
- Allow others to build off earlier work.
- Enable repurposing of materials and methods.

Reproducibility or Replicability?

1. You download data from another lab, following their methods, you try to regenerate their results. **Reproducibility**
2. You rerun the experiment that you ran in a previous paper but use a different mouse strain. **Replicability**
3. Someone performs your study using a cohort of subjects in Korea. **Replicability**
4. A colleague asks for your raw data and any scripts you may have to repeat your earlier analysis. **Reproducibility**

		Data	
		Same	Different
Code	Same	Reproducible	Replicable
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Different lab: same data & methods

Different mouse strain/cohort

Colleague uses your data & scripts

Kirstie Whitaker (doi: 10.6084/m9.figshare.5440621)

Threats to Reproducibility

Data & Methods

Data not public, incomplete methods, no raw data

Technical Rot

Operating system changes, URL rot, or software evolution.

Software Issues

Availability of software or "Custom Scripts" that aren't shared.

Missing Details

Protocols without enough detail or missing random seeds.

Threats to Replicability

Experimental Design

Poor design, overfitting, or selection bias.

Biological variation

Sex, age, genotype, or reagent differences.

Environment

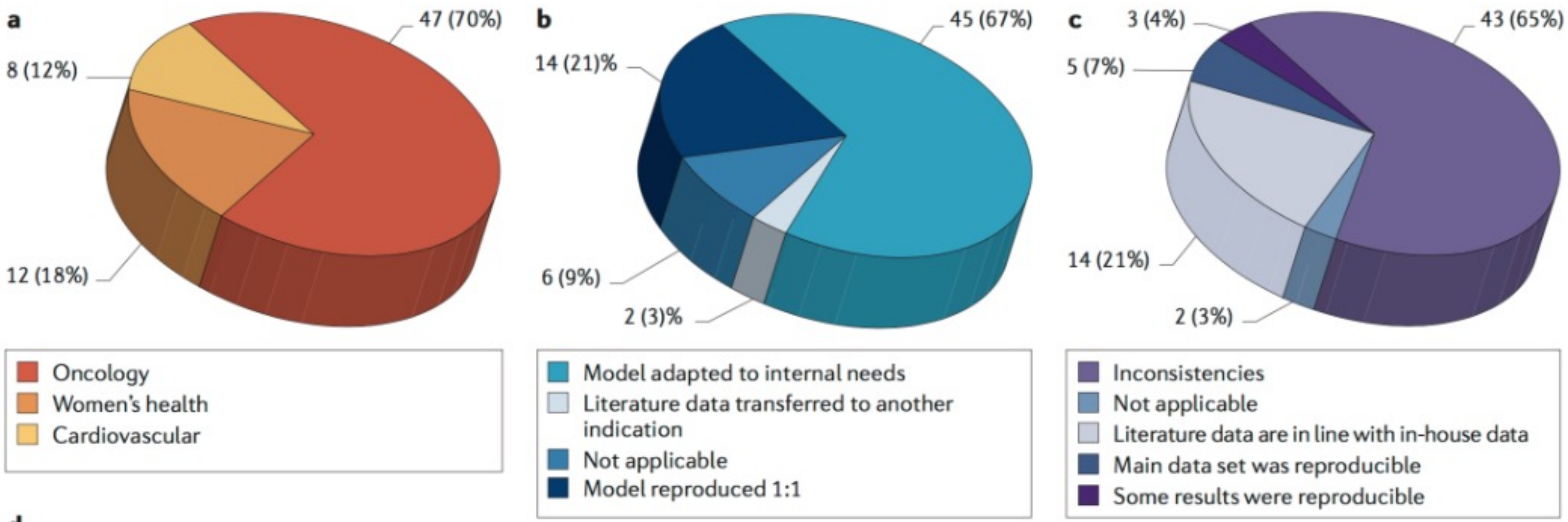
Storage conditions or confounding variables.

Integrity

Sloppiness or scientific misconduct.

The Reproducibility Crisis: Bayer

An unspoken rule among early-stage venture capital firms that "**at least 50% of published studies**, even those in top-tier academic journals, **can't be repeated with the same conclusions** by an industrial lab"



This is not good!

How difficult is achieving methodological reproducibility?

Paper Plane Exercise

1. Fold	Fold your airplane
2. Write	Write your instructions
3. Unfold	Unfold your airplane & pass both paper and instructions to your right
4. Reproduce	Reproduce your neighbor’s airplane

Is research that is reproducible and/or replicable necessarily correct?

Is a result that fails to reproduce and/or replicate someone's "fault"?

Reproducibility Best Practices

- Don’t assume prior knowledge
- Make raw data and scripts accessible.
- Document your software environment.
- **Use Version Control to track every step of the analysis pipeline.**

The Role of Version Control

Managing the "Code" and "Methods" pillars of reproducibility

The Enemy: "Filename Versioning"

Which one produced your Figure 1?

```
analysis_v1.py
analysis_v2_FINAL.py
analysis_v2_revised.py
analysis_FINAL_FOR_REAL.py
```

Manual tracking is prone to error and difficult to audit

What is Version Control?

Practice of **controlling**, **organizing**, and **tracking** different versions of computer files

- Allows you to revert to specific versions later
- Essential for software development, but useful for any evolving digital work (manuscripts, data, etc.)
- “Ctrl+Z” on steroids, with a complete history

Version Control with Git

- Software developed in 2005 by Linus Torvalds, the creator of Linux
- Widely used in the industry and academia
- Open source and free!

Git: A Research Time Machine

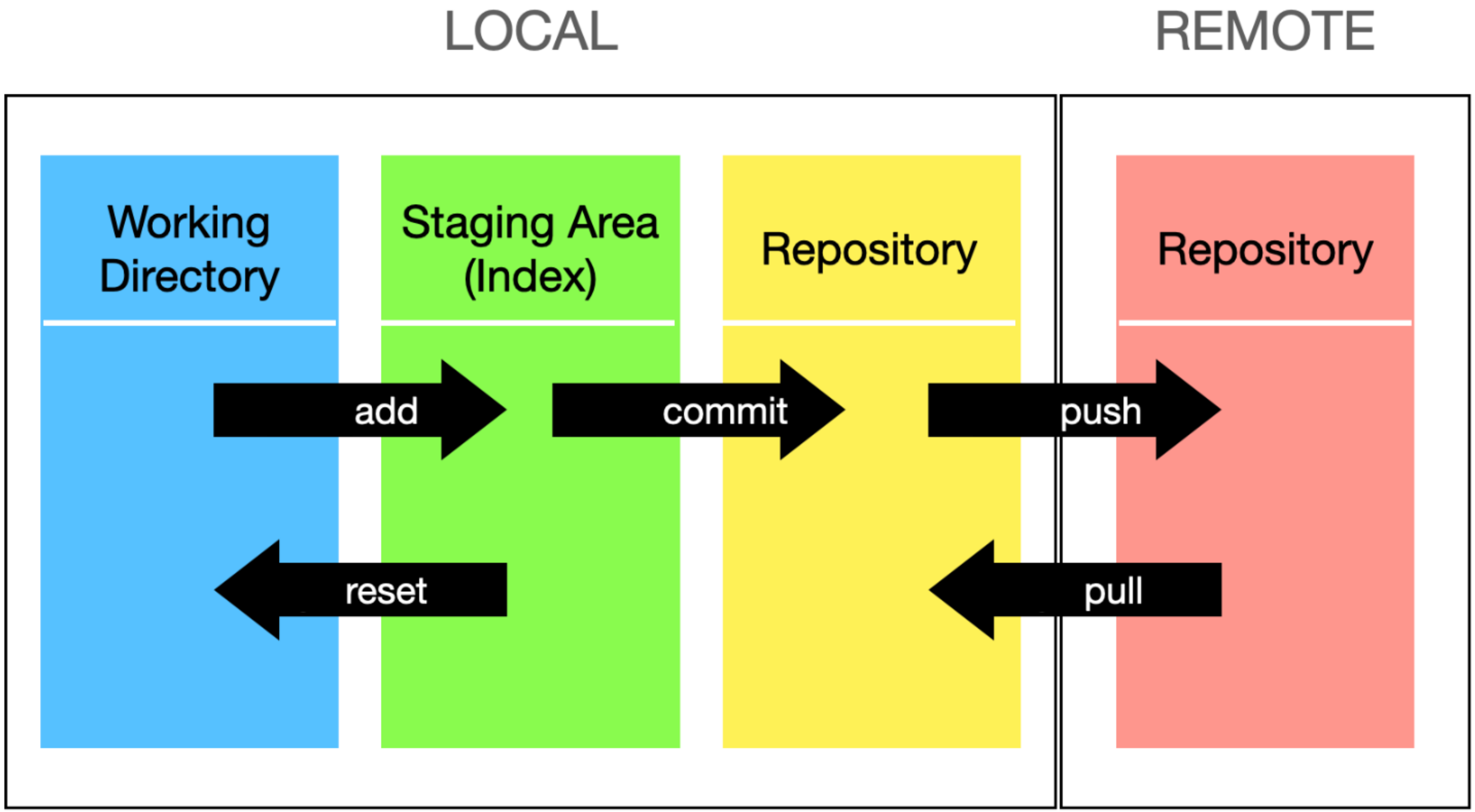
What is Git?

- **Version control system**
- **Tracks** every character change in your files.
- Allows you to **revert** to any previous working version instantly.
- Provides a **permanent record** of your analysis evolution.

How does Git do this?

- **Working Directory:** Local files you see and edit
- **Staging Area:** A temporary area to prepare changes for the next commit
- **Local Repository (.git directory):** The hidden folder where Git stores all the history and metadata for your project
- **Remote Repository:** A remote location where you can push your committed files for sharing, collaborating, showcasing skills

How do these structures interact?



The Power of the "Commit"

Snapshots
Captures the state of your entire project at once.

Attribution
Identifies who made which change → perfect for collaborations.

Documentation
Commit messages explain the 'why' behind each adjustment.

Integrity
Ensures history hasn't been tampered with or corrupted.

Today: Lesson 1

1. Create a new project & Initialize a local Git Repository
2. Stage and commit your files and changes
3. Review your version control history
4. Tell Git which files to ignore
5. Creating a new branch & merge

Resources Google Doc: **Setting up Git**

Getting started with Git

1. Start your small (e2-standard-4) instance & open JupyterLab.

2. Remove the old workshop repository (if present). In the **Terminal**, run:

```
rm -r dsc_workshop_2026/
```

3. Clone the workshop Git repository

GUI METHOD

1. In the top menu, choose **Git → Clone a Repository...**
2. Paste `https://github.com/ak-inbre-dsc/dsc_workshop_2026.git` and click **Clone**.
3. The repo appears in the file browser inside `~/dsc_workshop_2026/`.

TERMINAL ALTERNATIVE

```
cd ~
git clone https://github.com/ak-inbre-dsc/dsc_workshop_2026.git
```